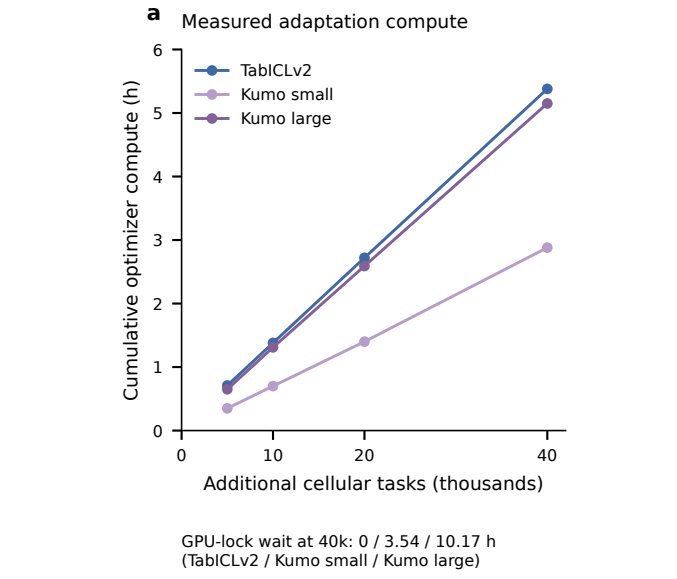
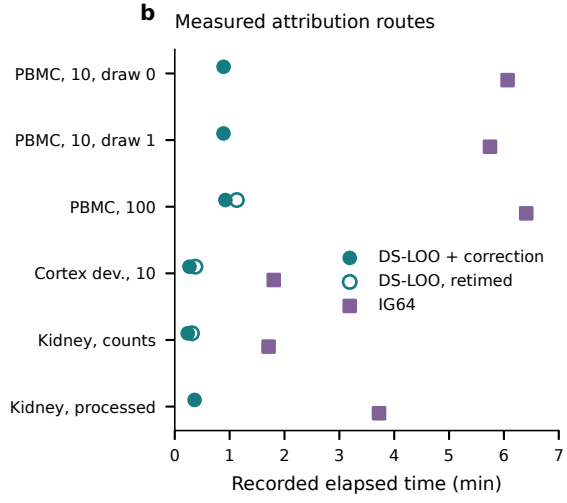
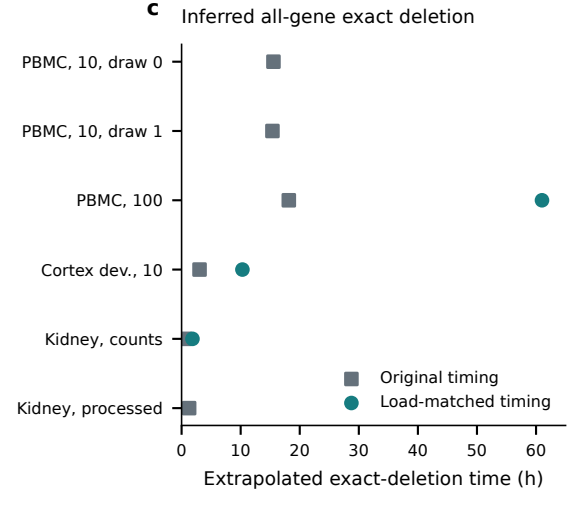
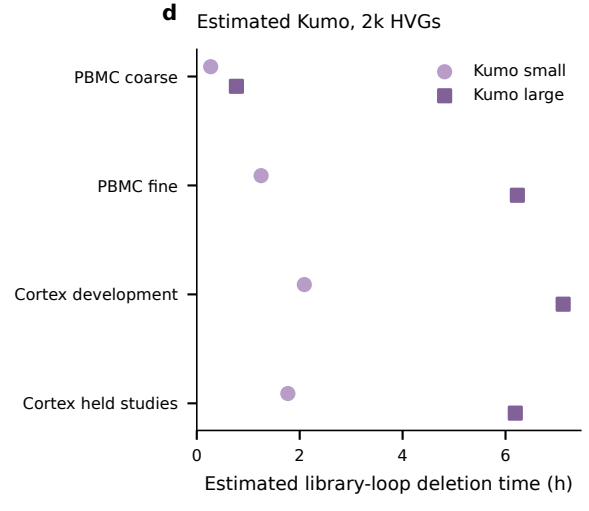
Recorded computational costs and timing boundaries









Measured (a,b), extrapolated (c), and estimated (d) quantities are shown separately. Timings are descriptive, not controlled speedups.
Attribution: fp32 CellPFN; Kumo estimates: fp16, different cell/gene scopes. No numerical simulator-throughput claim is made.